ONE-PAGE BRIEF

Post-quantum cryptography review: embit @ 2b375a33bd89

Client: embit (SeedSigner, Krux) — public repository (passive review) · Source: github.com/diybitcoinhardware/embit, commit 2b375a33bd89 · Reviewed: 2026-10-29 19:13 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY
Level 0 of 4
no post-quantum cryptography

FINDINGS
0 Critical **0** High **5** Medium
0 Low **1** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)
Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

Medium QSI-2026-06655	**Quantum-vulnerable ECDSA: transaction and message signing (39 file(s))** preliminary What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium QSI-2026-06656	**Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (7 file(s))** preliminary What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium QSI-2026-06658	**Quantum-vulnerable Pedersen commitment: threshold signing and MPC protocols (4 file(s))** preliminary What to do: Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the signature scheme; record which keys...
Medium QSI-2026-06657	**Quantum-vulnerable ECDSA: role not determined (4 file(s))** preliminary What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form...

1 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium threshold signing and MPC protocols	Pedersen commitment	Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the...
Medium transaction and message signing	ECDSA, Schnorr (BIP-340)	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium role not determined	ECDSA	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

Static review of source code and manifests; files analysed: 153. No requests were sent to the owner's systems.

Full report, content SHA-256: `6952d6f2effe767632633748509a85fe1bcc8364547c3efd53734d2d89571361`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.